# Element C

### Design Requirement #1 (HIGH Priority)- Preston

An underrated design requirement of our project is the handle. Including a handle on the dispenser's design would supply ease of transportation. The multiple boys' lacrosse coaches at Eleanor Roosevelt won't have to strain themselves by carrying ball bags and buckets. The handle would ideally reach a height near the stomach and hips, nearly removing the need to bend over completely. The wheels of the dispenser would be able to roll smoothly on multiple types of surfaces including dirt, concrete, sand, and even plastic turf. The handle would go hand-in-hand with the wheels, supplying the user with complete control over the dispenser.

### Design Requirement #2 (HIGH Priority)-Gabby

One important design requirement for our project is the button. The button mechanism will be connected to the spring-action dispenser. With the push of a button, or the kick of a button, our stakeholders will be able to dispense the balls individually and with maximum ease. The button is such a high priority because, without a lever or button mechanism, the balls would not eject from the container itself. Therefore, the button and the spring are arguably the most important requirements of the project.

### Design Requirement #3 (HIGH Priority) - Max

The materials used to create the ball launcher are a very high priority for us because it's the structure used to store and dispense the balls to the player. We need resources such as wood to make our structure in order to provide storage for the lacrosse balls as well as to use the wood to create a funnel for the balls to the springs that will propel the balls toward the lacrosse player. The reason why the materials used for our ball dispensers are so essential is so then our dispenser can be transportable to wherever the lacrosse team is practicing, our device being lightweight also makes it practical for daily use. Based on some prior research of different machines with the same concept of the lacrosse dispenser I was able to learn that other machines prioritized using plastics Instead as It allowed their device to be more compact and lightweight enough to carry. The type of materials we're using to create our device must be accounted for as we must make sure our device is compact enough to be stored for each lacrosse practice, we must also make sure that our device is durable enough to where it will not be broken easily due to repeat refilling and discharging of lacrosse balls. Taking Into account everything will cause our device to not only be efficient but durable, transportable, and beneficial for the ERHS lacrosse team to use.

## Design Requirement #4 (HIGH Priority)-Korni

The spring is a high priority for us because it is going to help our project work overall. We need the spring to make our design send out the balls to the players. The spring is related to our prior research because it is going to help propel our balls forward. We researched previously on different machines that do the same thing as our ball dispenser would. One of the machines that we researched was a basketball dispenser. The basketball dispenser we researched shoots balls out using an ejecting and retracting motion. This motion ensures that the ball is expelled out of the dispenser and then the dispenser grabs another ball to reject as well. This spring is going to be one of the most important components of our project because it is going to be what makes our design work. This spring is a vital part of our project because it is what begins the process of the dispenser. Another machine that inspired our dispenser is the cardboard gumball machine. The cardboard candy machine also gives us an idea of what we want to have in our ball dispenser design. Despite this being a small and compact design it allows us to be able to get a visual representation of what the inside of our dispenser can look like. We can tweak this for our personal design, and we can take the blueprint of it and develop it to make it our own.